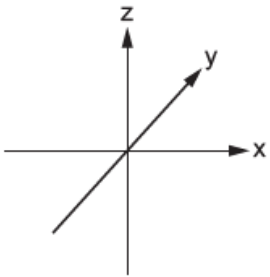
4 Haemoglobin is a critical protein found in red blood cells that carries oxygen from the lungs to the rest of the body.

Deoxyhaemoglobin and oxyhaemoglobin both contain iron in the +2 oxidation state. Each Fe^{2+} is coordinated to five nitrogen-containing ligands and one oxygen-containing ligand, forming an octahedral arrangement.

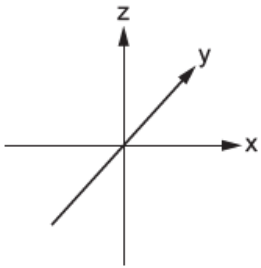
In an octahedral complex such as haemoglobin, the 3d subshell of Fe^{2+} is split into two energy levels.

(a) Using the axes in Fig. 4.1, draw **fully-labelled** diagrams of the following.

- One of the d orbitals at the lower energy level in an octahedral complex.
- One of the d orbitals at the higher energy level in an octahedral complex.



lower energy level



higher energy level

Fig. 4.1 **[2]**

(b) Haemoglobin can react with oxygen and carbon monoxide respectively as shown in the following two equilibria.

equilibrium 1 $Hb(aq) + 4O_2(aq) \rightleftharpoons Hb(O_2)_4(aq)$ $K_{c1} = 3 \times 10^{20}\ mol^{-4}\ dm^{12}$

equilibrium 2 $Hb(aq) + 4CO(aq) \rightleftharpoons Hb(CO)_4(aq)$ $K_{c2} = 6 \times 10^{22}\ mol^{-4}\ dm^{12}$

(i) Explain why carbon monoxide is toxic.

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.....

..... **[1]**

(ii) Carbon monoxide binds to haemoglobin, Hb, to form carboxyhaemoglobin, $Hb(CO)_4$.

If the percentage of haemoglobin bound to carbon monoxide reaches 45%, the result is fatal to humans.

Use the value of K_{c2} to calculate the concentration of carbon monoxide necessary for 45% of the Hb to be converted to $Hb(CO)_4$.

(iii) Equilibrium 1 and 2 can be expressed as a single equilibrium 3.

equilibrium 3 $Hb(CO)_4(aq) + 4O_2(aq) \rightleftharpoons Hb(O_2)_4(aq) + 4CO(aq)$

Using K_{c1} and K_{c2}, calculate the value of K_c for equilibrium 3.

[1]

(iv) Use the K_C value calculated in **b(iii)** to suggest the position of equilibrium and the sign for ΔG for equilibrium 3.

.....

..... **[1]**

(v) A patient suffering from carbon monoxide poisoning can be treated by giving pure oxygen to breathe. Suggest a reason why this treatment is effective.

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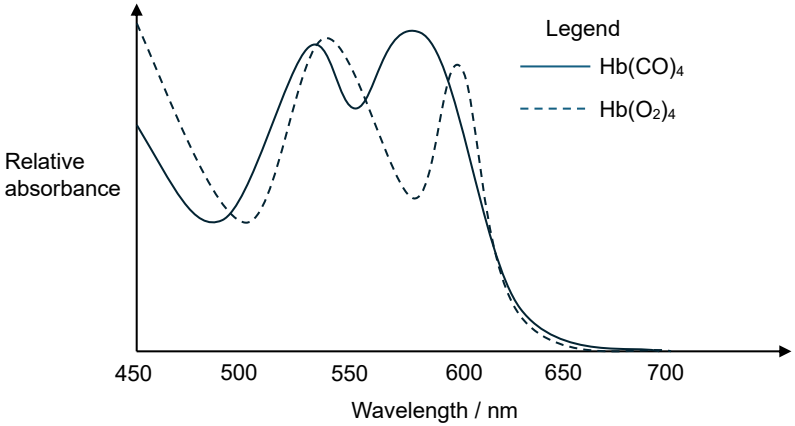
..... **[2]**

(c) Carboxyhaemoglobin, $Hb(CO)_4$, and oxyhaemoglobin, $Hb(O_2)_4$, were analysed and the absorption spectrum was observed.



Species	Colour Observed
$Hb(CO)_4$	Cherry-red
$Hb(O_2)_4$	Orange-red

Colour	Wavelength (nm)	Colour	Wavelength (nm)
Violet	380 – 400	Yellow	560 – 580
Blue	400 – 490	Orange	580 – 620
Green	490 – 560	Red	620 – 800

(i) With reference to the absorption spectrum, explain why both $Hb(CO)_4$ and $Hb(O_2)_4$ are generally red in colour.

.....

..... **[1]**

(ii) Suggest why $Hb(CO)_4$ and $Hb(O_2)_4$ have different shades of red.

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..... **[2]**

[Total: 12]

5 Cycling is a demanding endurance sport that pushes athletes to optimise every aspect of